Name: _____

TOPIC TEST 2

Chapter 3: Functions and sketching graphs

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 8 short-answer questions (16 marks).
- Part C: 2 analysis questions (6 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 What is the composite function $f(g(x))$ given that $f(x) = \sin(x)$ and $g(x) = 2x$?

A $2\sin(x)$

B $2x$

C $2\sin(2x)$

D $\sin(2x)$

E $2x\sin(2x)$

[1 mark]

2 A function can be identified as one-to-one if, for all values in the domain,

A $\frac{dy}{dx} = 0$.

B $\frac{dy}{dx} > 0$.

C $\frac{d^2y}{dx^2} > 0$.

D $\frac{d^2y}{dx^2} < 0$.

E $\frac{d^2y}{dx^2} \leq 0$.

[1 mark]

3 Write the inverse function of $y = 2x + 3$.

A $y = 2x - 3$

B $y = \frac{1}{2}x + 3$

C $y = \frac{x + 3}{2}$

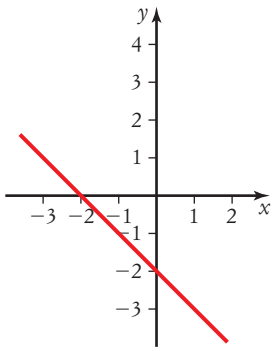
D $y = \frac{x - 3}{2}$

E $y = x - 3$

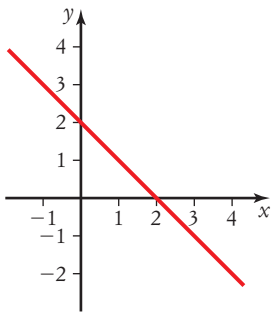
[1 mark]

4 Sketch the function $y = -|x| + 2$.

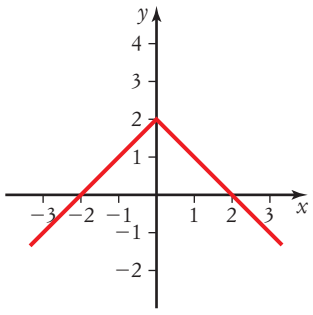
A



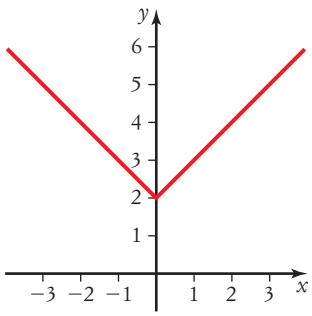
B



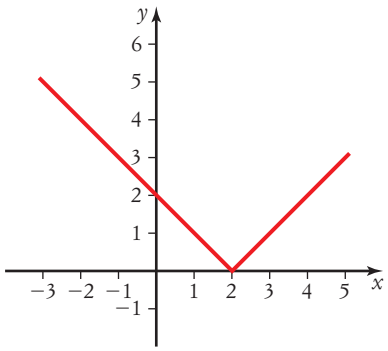
C



D



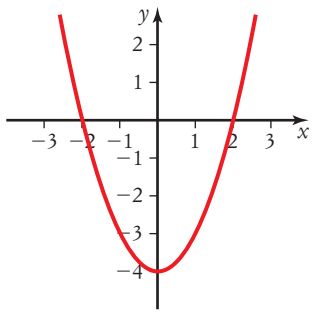
E



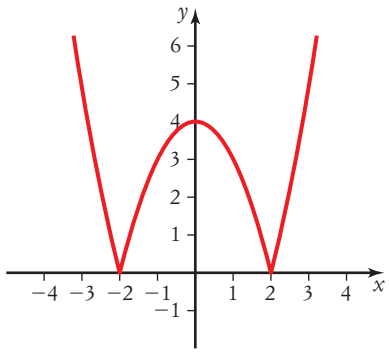
[1 mark]

5 Which graph represents the reciprocal function of $f(x) = (x + 2)(x - 2)$?

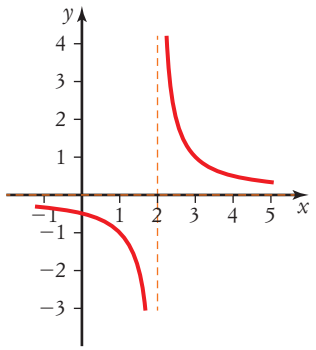
A



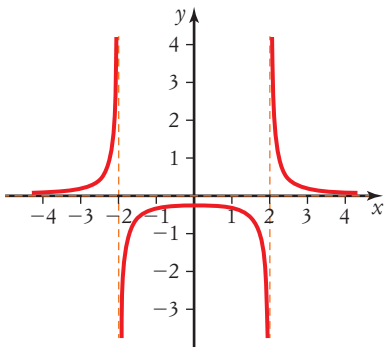
B



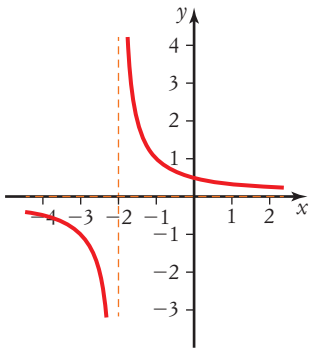
C



D



E



[1 mark]

Part B

Short-answer questions

8 short-answer questions
16 marks
Show your working where appropriate.

1 Sketch the graph of $y = f(|x|)$ given that $f(x) = \ln(-x)$.

[2 marks]

2 Sketch the graph of $y = \dfrac{1}{f(x)}$ given that $f(x) = \ln(-x)$. Write the domain of $f(x)$.

[2 marks]

3 Sketch the graph of $f(x) = |x - 1|$. Write the range of $f(x)$.

[2 marks]

4 Sketch the graph of $f(x) = \dfrac{1}{|x - 1|}$. Write the domain of $f(x)$.

[2 marks]

5 Sketch the graph of $f(x) = \dfrac{x}{|x - 1|}$. Write the range of $f(x)$.

[2 marks]

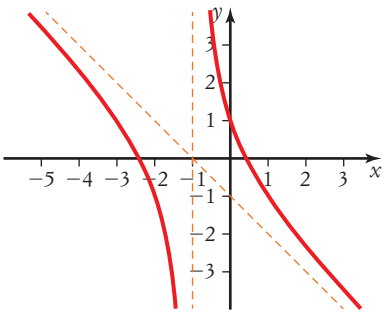
6 Sketch the graph of $f(x) = \dfrac{x^2}{|x - 1|}$. Write the domain of $f(x)$.

[2 marks]

7 Sketch the graph of $f(x) = \dfrac{x^2 - 2x + 3}{x - 1}$.

[2 marks]

8 Write a rational function for the following graph.



[2 marks]

Part C

Analysis questions

2 analysis questions
6 marks
Show your working where appropriate.

1 Solve the inequality $\frac{2x+1}{x-2} > 3$, by sketching functions.

[3 marks]

2 A university environmental team are studying the cost to remove contaminated soil. The function $C = \frac{15p}{1.1-p}$ models the estimated cost (C in 1000s) to remove the percentage (p as a decimal) of the contaminated soil.

a State a reasonable domain and range for the function.

b Use a graph of the function to find the percentage of the contaminated soil that could be removed on a budget of $75 000.

[3 marks]

[Total marks: 27]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $f(g(x)) = \sin(2x)$,

$\therefore$ **D** **[1 mark]**

2 Increasing $\frac{dy}{dx} > 0$,

$\therefore$ **B** **[1 mark]**

3 $y = \frac{x-3}{2}$,

$\therefore$ **D** **[1 mark]**

4 $y = |x|$ is inverted and moved up 2 to make $y = -|x| + 2$.

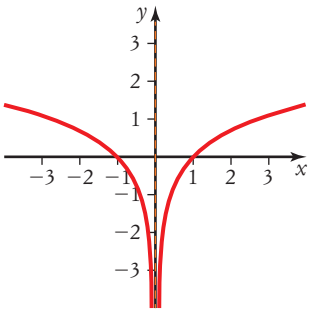
$\therefore$ **C** **[1 mark]**

5 $y = \frac{1}{(x-2)(x+2)}$,

$\therefore$ **D** **[1 mark]**

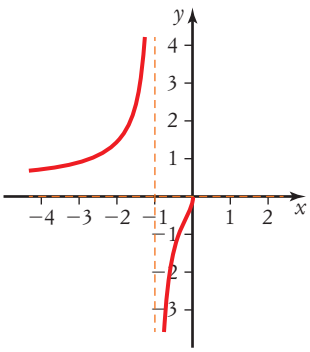
Part B – Short-answer questions

1 Graph:



[2 marks]

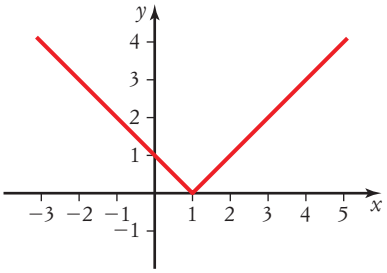
2 Domain: $x < 0, x \neq -1$ **[1 mark]**



[1 mark]

[2 marks altogether]

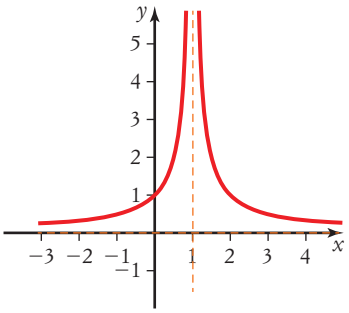
3 Range: $y \geq 0$ **[1 mark]**



[1 mark]

[2 marks altogether]

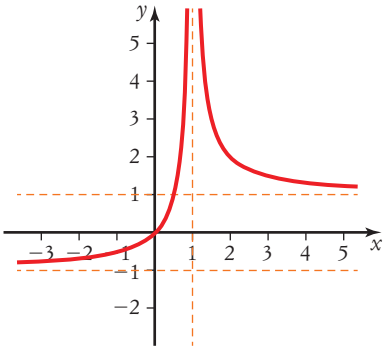
4 Domain: $x \in \mathbf{R}, x \neq 1$ **[1 mark]**



[1 mark]

[2 marks altogether]

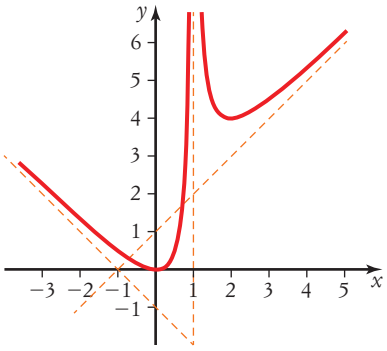
5 Range: $y > -1$ **[1 mark]**



[1 mark]

[2 marks altogether]

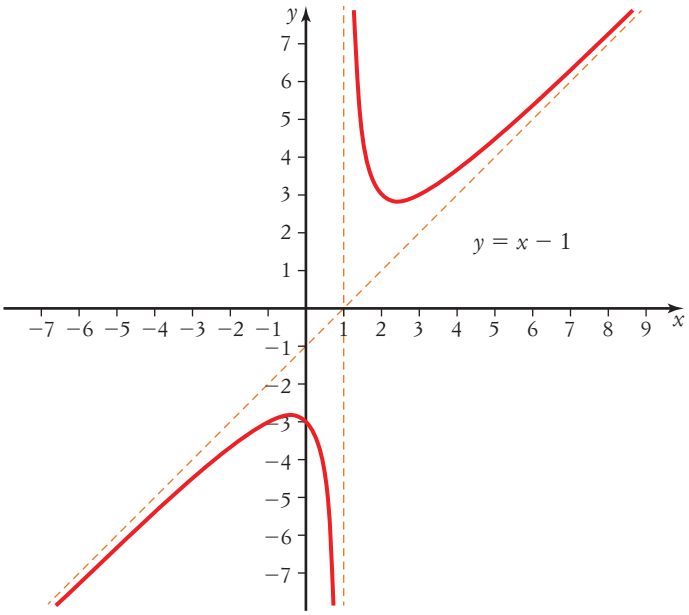
6 Domain: $x \in \mathbf{R}, x \neq 1$ **[1 mark]**



[1 mark]

[2 marks altogether]

7 Asymptotes **[1 mark]**



[1 mark]

[2 marks altogether]

8 Correct denominator **[1 mark]**

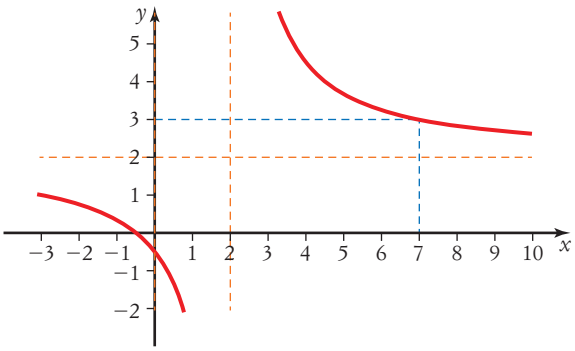
$$y = \frac{1 - 2x - x^2}{x + 1}$$ **[1 mark]**

[2 marks altogether]

Part C – Analysis questions

1 $2x + 1 > 3(x - 2)$, **[1 mark]**

$x < 7$



[1 mark]

Solution is $2 < x < 7$ **[1 mark]**

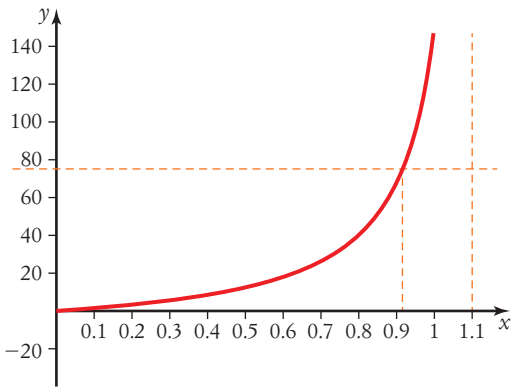
[3 marks altogether]

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2 a Domain: $0 \leq p \leq 1$,

Range: $0 \leq C \leq 150000$ **[1 mark]**

b



[1 mark]

0.92 or 92% of contaminated soil can be removed. **[1 mark]**

[3 marks altogether]